

Supplementary Appendix — Tables S1 to S21

Paper: Olfactory dysfunction, daytime activity reduction, and 24-hour rhythm fragmentation — but not food-odor recognition — in NHANES 2013–2014

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Notes on formatting

- **β** (beta) values are OD effects from M4-adjusted survey-weighted regression unless otherwise stated.
- **95% CIs** are Wald (or cluster-bootstrap where indicated).
- **p-values** are uncorrected; **q-values** are Benjamini-Hochberg FDR-corrected within each test family.
- **Cohen's d** is standardized by the survey-weighted SD of the outcome.
- Significance markers: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
- Missing/non-estimable cells: '—'.

Table S1. M4 model — complete coefficient table

All covariate coefficients (age, sex, race, education, PIR, BMI, smoking, diabetes, comorbidity, PHQ-9, sinusitis, head injury, medications) for each primary outcome. Coefficients are on the native scale of each outcome.

Outcome	Term	β	95% CI	p
Mean MIMS/day	(Intercept)	21.03	(11.91, 30.15)	0.022
Mean MIMS/day	OD (binary)	-0.612	(-0.815, -0.409)	0.017
Mean MIMS/day	Age	-0.084	(-0.130, -0.038)	0.027
Mean MIMS/day	Female	1.153	(-2.592, 4.898)	0.159
Mean MIMS/day	Mexican American	1.221	(-5.558, 7.999)	0.262
Mean MIMS/day	Non-Hispanic Asian	-0.262	(-0.516, -0.0079)	0.048
Mean MIMS/day	Non-Hispanic Black	-0.557	(-4.724, 3.610)	0.339
Mean MIMS/day	Other Hispanic	1.38	(0.833, 1.927)	0.02
Mean MIMS/day	Other/Multi	-0.856	(-4.372, 2.661)	0.199
Mean MIMS/day	Education: HS	1.086	(-2.197, 4.369)	0.149
Mean MIMS/day	Education: some college	0.616	(-1.414, 2.647)	0.161
Mean MIMS/day	Education: < HS	1.177	(-3.618, 5.971)	0.198
Mean MIMS/day	PIR	-0.0091	(-1.541, 1.523)	0.952
Mean MIMS/day	BMI	-0.096	(-0.115, -0.077)	0.01
Mean MIMS/day	Smoker: current	-0.878	(-1.394, -0.363)	0.029
Mean MIMS/day	Smoker: former	0.0066	(-2.031, 2.044)	0.974
Mean MIMS/day	Diabetes (self-report)	-0.833	(-6.897, 5.232)	0.331
Mean MIMS/day	Comorbidity count	-0.114	(-0.380, 0.151)	0.115
Mean MIMS/day	PHQ-9 score	-0.008	(-0.316, 0.300)	0.796
Mean MIMS/day	Sinusitis (ever)	0.208	(-7.132, 7.548)	0.78
Mean MIMS/day	Head injury	-0.259	(-3.073, 2.556)	0.451

Outcome	Term	β	95% CI	p
Mean MIMS/day	N medications	-0.069	(-0.993, 0.855)	0.518
Sedentary min/day	(Intercept)	311	(187.6, 434.3)	0.02
Sedentary min/day	OD (binary)	-2.934	(-156.6, 150.7)	0.848
Sedentary min/day	Age	1.142	(-0.656, 2.940)	0.078
Sedentary min/day	Female	-33.11	(-114.0, 47.74)	0.121
Sedentary min/day	Mexican American	-17.92	(-144.0, 108.1)	0.322
Sedentary min/day	Non-Hispanic Asian	17.05	(-68.06, 102.2)	0.238
Sedentary min/day	Non-Hispanic Black	22.57	(-50.77, 95.91)	0.159
Sedentary min/day	Other Hispanic	-19.04	(-103.1, 64.98)	0.213
Sedentary min/day	Other/Multi	29.9	(-140.9, 200.7)	0.269
Sedentary min/day	Education: HS	-40.72	(-119.8, 38.35)	0.097
Sedentary min/day	Education: some college	-25.33	(-71.02, 20.36)	0.09
Sedentary min/day	Education: < HS	-37.05	(-101.5, 27.38)	0.087
Sedentary min/day	PIR	3.245	(-21.69, 28.18)	0.346
Sedentary min/day	BMI	1.938	(1.265, 2.610)	0.017
Sedentary min/day	Smoker: current	16.32	(-14.01, 46.65)	0.092
Sedentary min/day	Smoker: former	-8.169	(-53.89, 37.55)	0.264
Sedentary min/day	Diabetes (self-report)	20.46	(-74.39, 115.3)	0.223
Sedentary min/day	Comorbidity count	4.006	(-11.79, 19.80)	0.192
Sedentary min/day	PHQ-9 score	-0.054	(-12.89, 12.78)	0.966
Sedentary min/day	Sinusitis (ever)	-4.365	(-228.1, 219.4)	0.845
Sedentary min/day	Head injury	6.767	(-26.67, 40.21)	0.236
Sedentary min/day	N medications	-1.046	(-39.04, 36.94)	0.786
MVPA min/day	(Intercept)	109.7	(48.11, 171.3)	0.028
MVPA min/day	OD (binary)	-3.44	(-37.24, 30.36)	0.419
MVPA min/day	Age	-0.996	(-1.432, -0.561)	0.022
MVPA min/day	Female	7.696	(-11.22, 26.61)	0.122
MVPA min/day	Mexican American	15.32	(-71.17, 101.8)	0.266
MVPA min/day	Non-Hispanic Asian	0.36	(-31.43, 32.15)	0.909
MVPA min/day	Non-Hispanic Black	-6.047	(-14.11, 2.016)	0.067
MVPA min/day	Other Hispanic	17.79	(-12.42, 48.01)	0.085
MVPA min/day	Other/Multi	-6.233	(-47.78, 35.31)	0.308
MVPA min/day	Education: HS	5.646	(-21.37, 32.66)	0.229
MVPA min/day	Education: some college	1.453	(-26.86, 29.77)	0.632
MVPA min/day	Education: < HS	6.097	(-30.79, 42.98)	0.283
MVPA min/day	PIR	-0.967	(-13.79, 11.86)	0.514
MVPA min/day	BMI	-0.784	(-1.320, -0.247)	0.034
MVPA min/day	Smoker: current	-8.456	(-18.28, 1.372)	0.058
MVPA min/day	Smoker: former	2.322	(-9.535, 14.18)	0.243

Outcome	Term	β	95% CI	p
MVPA min/day	Diabetes (self-report)	-8.238	(-62.88, 46.41)	0.306
MVPA min/day	Comorbidity count	-0.151	(-8.088, 7.787)	0.849
MVPA min/day	PHQ-9 score	-0.229	(-0.868, 0.409)	0.137
MVPA min/day	Sinusitis (ever)	0.617	(-53.82, 55.05)	0.909
MVPA min/day	Head injury	0.177	(-23.58, 23.94)	0.94
MVPA min/day	N medications	-0.375	(-9.994, 9.243)	0.707
IS	(Intercept)	0.538	(0.492, 0.584)	0.004
IS	OD (binary)	-0.028	(-0.057, 0.0008)	0.051
IS	Age	0.0022	(0.0012, 0.0032)	0.022
IS	Female	0.03	(-0.138, 0.198)	0.264
IS	Mexican American	0.01	(-0.227, 0.248)	0.684
IS	Non-Hispanic Asian	-0.015	(-0.253, 0.224)	0.574
IS	Non-Hispanic Black	-0.042	(-0.251, 0.166)	0.236
IS	Other Hispanic	-0.0003	(-0.013, 0.012)	0.84
IS	Other/Multi	-0.034	(-0.324, 0.255)	0.373
IS	Education: HS	0.032	(-0.132, 0.196)	0.244
IS	Education: some college	0.017	(-0.113, 0.148)	0.339
IS	Education: < HS	0.057	(-0.158, 0.271)	0.185
IS	PIR	0.0012	(-0.012, 0.014)	0.467
IS	BMI	-0.0029	(-0.0064, 0.0006)	0.061
IS	Smoker: current	-0.03	(-0.232, 0.173)	0.313
IS	Smoker: former	0.013	(-0.020, 0.046)	0.123
IS	Diabetes (self-report)	-0.026	(-0.176, 0.125)	0.272
IS	Comorbidity count	-0.0057	(-0.043, 0.032)	0.301
IS	PHQ-9 score	-0.0027	(-0.011, 0.0057)	0.154
IS	Sinusitis (ever)	-0.019	(-0.292, 0.253)	0.534
IS	Head injury	0.0041	(-0.269, 0.277)	0.879
IS	N medications	-0.0009	(-0.033, 0.031)	0.773
IV	(Intercept)	0.518	(0.412, 0.624)	0.01
IV	OD (binary)	0.048	(-0.273, 0.368)	0.311
IV	Age	0.0022	(-0.0037, 0.0082)	0.13
IV	Female	-0.015	(-0.132, 0.102)	0.351
IV	Mexican American	-0.034	(-0.508, 0.439)	0.524
IV	Non-Hispanic Asian	0.047	(0.039, 0.055)	0.008
IV	Non-Hispanic Black	0.029	(-0.410, 0.468)	0.56
IV	Other Hispanic	-0.026	(-0.437, 0.386)	0.572
IV	Other/Multi	0.027	(-0.417, 0.471)	0.579
IV	Education: HS	-0.098	(-0.369, 0.173)	0.137
IV	Education: some college	-0.06	(-0.234, 0.114)	0.144

Outcome	Term	β	95% CI	p
IV	Education: < HS	-0.116	(-0.272, 0.041)	0.067
IV	PIR	0.0036	(-0.075, 0.082)	0.666
IV	BMI	0.0031	(-0.0010, 0.0072)	0.065
IV	Smoker: current	-0.031	(-0.147, 0.086)	0.185
IV	Smoker: former	-0.03	(-0.082, 0.022)	0.086
IV	Diabetes (self-report)	0.028	(-0.073, 0.129)	0.174
IV	Comorbidity count	0.014	(-0.120, 0.148)	0.407
IV	PHQ-9 score	0.0019	(-0.0010, 0.0049)	0.076
IV	Sinusitis (ever)	0.027	(-0.241, 0.294)	0.424
IV	Head injury	0.018	(-0.299, 0.335)	0.605
IV	N medications	0.0034	(-0.029, 0.036)	0.416
RA	(Intercept)	1.013	(0.968, 1.058)	0.002
RA	OD (binary)	-0.0067	(-0.079, 0.066)	0.45
RA	Age	-0.0006	(-0.0007, -0.0006)	0.002
RA	Female	0.0073	(0.0053, 0.0093)	0.013
RA	Mexican American	0.0032	(-0.038, 0.044)	0.502
RA	Non-Hispanic Asian	-0.0098	(-0.058, 0.039)	0.236
RA	Non-Hispanic Black	-0.011	(-0.054, 0.032)	0.186
RA	Other Hispanic	-0.0019	(-0.023, 0.019)	0.453
RA	Other/Multi	-0.011	(-0.162, 0.141)	0.536
RA	Education: HS	0.0023	(-0.0099, 0.015)	0.25
RA	Education: some college	0.0032	(-0.012, 0.018)	0.224
RA	Education: < HS	0.0007	(-0.047, 0.048)	0.885
RA	PIR	0.0017	(-0.0004, 0.0038)	0.06
RA	BMI	-0.0009	(-0.0025, 0.0008)	0.096
RA	Smoker: current	-0.0061	(-0.073, 0.060)	0.454
RA	Smoker: former	-0.0024	(-0.069, 0.065)	0.728
RA	Diabetes (self-report)	-0.0067	(-0.026, 0.013)	0.142
RA	Comorbidity count	-0.0007	(-0.011, 0.010)	0.574
RA	PHQ-9 score	-0.0002	(-0.0050, 0.0046)	0.675
RA	Sinusitis (ever)	-0.0079	(-0.095, 0.079)	0.451
RA	Head injury	-0.0023	(-0.032, 0.027)	0.503
RA	N medications	0.0002	(-0.0037, 0.0041)	0.608
Total sleep (min)	(Intercept)	377.5	(60.70, 694.3)	0.042
Total sleep (min)	OD (binary)	-2.753	(-12.73, 7.220)	0.177
Total sleep (min)	Age	-0.375	(-3.622, 2.873)	0.381
Total sleep (min)	Female	4.638	(-106.1, 115.4)	0.689
Total sleep (min)	Mexican American	-9.829	(-28.16, 8.505)	0.093
Total sleep (min)	Non-Hispanic Asian	-22.29	(-99.90, 55.31)	0.17

Outcome	Term	β	95% CI	p
Total sleep (min)	Non-Hispanic Black	-12.56	(-18.16, -6.962)	0.022
Total sleep (min)	Other Hispanic	-15.51	(-25.56, -5.454)	0.032
Total sleep (min)	Other/Multi	-24.84	(-131.3, 81.65)	0.207
Total sleep (min)	Education: HS	8.872	(-0.229, 17.97)	0.051
Total sleep (min)	Education: some college	9.714	(-59.61, 79.04)	0.326
Total sleep (min)	Education: < HS	6.173	(-1.137, 13.48)	0.059
Total sleep (min)	PIR	3.771	(-15.26, 22.80)	0.241
Total sleep (min)	BMI	-0.662	(-1.471, 0.148)	0.061
Total sleep (min)	Smoker: current	-11.93	(-75.33, 51.47)	0.252
Total sleep (min)	Smoker: former	-5.323	(-90.79, 80.14)	0.574
Total sleep (min)	Diabetes (self-report)	1.13	(-14.16, 16.42)	0.52
Total sleep (min)	Comorbidity count	-1.304	(-59.64, 57.03)	0.824
Total sleep (min)	PHQ-9 score	-0.31	(-5.884, 5.264)	0.609
Total sleep (min)	Sinusitis (ever)	-8.167	(-56.02, 39.69)	0.275
Total sleep (min)	Head injury	5.804	(-72.63, 84.24)	0.52
Total sleep (min)	N medications	-0.938	(-7.474, 5.599)	0.319
Sleep efficiency	(Intercept)	0.924	(0.895, 0.953)	0.002
Sleep efficiency	OD (binary)	-0.0059	(-0.051, 0.039)	0.345
Sleep efficiency	Age	-0.0003	(-0.0020, 0.0014)	0.268
Sleep efficiency	Female	0.0063	(-0.0005, 0.013)	0.054
Sleep efficiency	Mexican American	-0.0028	(-0.080, 0.075)	0.723
Sleep efficiency	Non-Hispanic Asian	-0.013	(-0.025, -0.0005)	0.048
Sleep efficiency	Non-Hispanic Black	-0.0093	(-0.022, 0.0030)	0.066
Sleep efficiency	Other Hispanic	-0.0052	(-0.043, 0.033)	0.333
Sleep efficiency	Other/Multi	-0.0042	(-0.126, 0.117)	0.737
Sleep efficiency	Education: HS	0.0023	(-0.065, 0.070)	0.743
Sleep efficiency	Education: some college	0.0026	(-0.078, 0.083)	0.749
Sleep efficiency	Education: < HS	-0.0026	(-0.027, 0.022)	0.408
Sleep efficiency	PIR	0.0027	(-0.0014, 0.0069)	0.076
Sleep efficiency	BMI	-0.0004	(-0.0033, 0.0025)	0.333
Sleep efficiency	Smoker: current	-0.0069	(-0.075, 0.061)	0.42
Sleep efficiency	Smoker: former	-0.0006	(-0.051, 0.050)	0.908
Sleep efficiency	Diabetes (self-report)	-0.0053	(-0.015, 0.0041)	0.088
Sleep efficiency	Comorbidity count	0.0011	(-0.0061, 0.0082)	0.309
Sleep efficiency	PHQ-9 score	-0.0005	(-0.0038, 0.0028)	0.312
Sleep efficiency	Sinusitis (ever)	-0.0077	(-0.028, 0.012)	0.13
Sleep efficiency	Head injury	-0.0028	(-0.012, 0.0063)	0.16
Sleep efficiency	N medications	-0.0008	(-0.0035, 0.0019)	0.169
WASO (min)	(Intercept)	31.2	(-24.50, 86.89)	0.089

Outcome	Term	β	95% CI	p
WASO (min)	OD (binary)	2.219	(-17.58, 22.02)	0.39
WASO (min)	Age	0.082	(-1.008, 1.171)	0.516
WASO (min)	Female	-1.569	(-16.30, 13.16)	0.405
WASO (min)	Mexican American	-0.054	(-27.82, 27.71)	0.984
WASO (min)	Non-Hispanic Asian	2.637	(-6.343, 11.62)	0.167
WASO (min)	Non-Hispanic Black	2.633	(1.775, 3.491)	0.016
WASO (min)	Other Hispanic	-0.046	(-25.35, 25.26)	0.985
WASO (min)	Other/Multi	-1.437	(-30.13, 27.26)	0.639
WASO (min)	Education: HS	0.2	(-29.34, 29.74)	0.945
WASO (min)	Education: some college	0.062	(-23.66, 23.79)	0.979
WASO (min)	Education: < HS	1.943	(-7.416, 11.30)	0.231
WASO (min)	PIR	-0.633	(-1.578, 0.312)	0.074
WASO (min)	BMI	0.107	(-0.826, 1.041)	0.382
WASO (min)	Smoker: current	1.731	(-19.31, 22.77)	0.486
WASO (min)	Smoker: former	-0.337	(-12.53, 11.85)	0.785
WASO (min)	Diabetes (self-report)	3.026	(-2.919, 8.972)	0.098
WASO (min)	Comorbidity count	-0.458	(-6.213, 5.296)	0.496
WASO (min)	PHQ-9 score	0.17	(-0.855, 1.196)	0.282
WASO (min)	Sinusitis (ever)	1.425	(-7.806, 10.66)	0.3
WASO (min)	Head injury	2.191	(-2.029, 6.412)	0.096
WASO (min)	N medications	0.019	(-0.663, 0.701)	0.784
ASTP	(Intercept)	0.151	(-0.099, 0.401)	0.082
ASTP	OD (binary)	0.011	(0.0009, 0.022)	0.046
ASTP	Age	0.0005	(-0.0006, 0.0016)	0.109
ASTP	Female	-0.023	(-0.126, 0.080)	0.215
ASTP	Mexican American	-0.024	(-0.135, 0.087)	0.222
ASTP	Non-Hispanic Asian	-0.0044	(-0.022, 0.013)	0.195
ASTP	Non-Hispanic Black	0.004	(-0.110, 0.118)	0.735
ASTP	Other Hispanic	-0.024	(-0.056, 0.0085)	0.068
ASTP	Other/Multi	0.012	(-0.018, 0.041)	0.124
ASTP	Education: HS	-0.023	(-0.126, 0.080)	0.215
ASTP	Education: some college	-0.013	(-0.040, 0.015)	0.105
ASTP	Education: < HS	-0.024	(-0.098, 0.051)	0.155
ASTP	PIR	-5.14e-05	(-0.036, 0.036)	0.988
ASTP	BMI	0.0019	(0.0002, 0.0035)	0.044
ASTP	Smoker: current	0.019	(-0.0088, 0.046)	0.074
ASTP	Smoker: former	0.0008	(-0.052, 0.053)	0.879
ASTP	Diabetes (self-report)	0.014	(-0.087, 0.115)	0.331
ASTP	Comorbidity count	0.0038	(-0.010, 0.018)	0.18

Outcome	Term	β	95% CI	p
ASTP	PHQ-9 score	0.0001	(-0.0083, 0.0086)	0.859
ASTP	Sinusitis (ever)	-0.0042	(-0.101, 0.092)	0.681
ASTP	Head injury	-0.0017	(-0.042, 0.039)	0.694
ASTP	N medications	0.0017	(-0.015, 0.018)	0.418

Table S2. M4 primary outcomes with BH-FDR-adjusted q-values

Five pre-specified primary outcomes from the fully adjusted M4 model. p-values are uncorrected; q-values are Benjamini-Hochberg FDR-corrected within each aim family. n = 2,055 or 1,683 per outcome.

Aim	Outcome	β	95% CI	p	q (FDR)
Aim 1 — Mean daily MIMS (PRIMARY)	Mean MIMS/day	-0.612	(-0.930, -0.294)	<.001	0.001
Aim 1 — Sedentary min/day	Sedentary min/day	-2.934	(-16.56, 10.69)	.663	0.663
Aim 1 — MVPA min/day	MVPA min/day	-3.44	(-6.449, -0.431)	.026	0.04
Aim 2 — Interdaily stability	IS	-0.028	(-0.053, -0.0040)	.024	0.036
Aim 2 — Intradaily variability	IV	0.048	(0.011, 0.084)	.012	0.036
Aim 2 — Relative amplitude	RA	-0.0067	(-0.015, 0.0018)	.117	0.117
Aim 3 — Total sleep time (min)	Total sleep (min)	-2.753	(-12.74, 7.235)	.577	0.577
Aim 3 — Sleep efficiency	Sleep efficiency	-0.0059	(-0.015, 0.0036)	.215	0.287
Aim 3 — WASO (min)	WASO (min)	2.219	(-1.211, 5.649)	.196	0.287
Aim 3 — ASTP	ASTP	0.011	(0.0030, 0.020)	.009	0.036

Table S3. E-values for unmeasured confounding

E-values for the OD effect point estimate and for the lower bound of the 95% CI, per VanderWeele & Ding (2017). Larger E-values indicate greater robustness to unmeasured confounding.

Outcome	β (95% CI)	p	Cohen's d	E-value (point)	E-value (CI)
Mean MIMS/day	-0.612 (-0.930, -0.294)	0	-0.177	1.63	1.38
Sedentary min/day	-2.934 (-16.556, +10.689)	0.663	-0.031	1.2	1
MVPA min/day	-3.440 (-6.449, -0.431)	0.026	-0.096	1.41	1.12
IS	-0.028 (-0.053, -0.004)	0.024	-0.203	1.7	1.19
IV	+0.048 (+0.011, +0.084)	0.012	0.208	1.71	1.26
RA	-0.007 (-0.015, +0.002)	0.117	-0.153	1.56	1
Total sleep (min)	-2.753 (-12.741, +7.235)	0.577	-0.038	1.22	1
Sleep efficiency	-0.006 (-0.015, +0.004)	0.215	-0.119	1.47	1
WASO (min)	+2.219 (-1.211, +5.649)	0.196	0.114	1.46	1
ASTP	+0.011 (+0.003, +0.020)	0.00	0.157	1.57	1.24

Table S4. OD severity gradient analysis

Effect of severity-gradient OD definitions. Left columns: continuous PST score 0–8 (each additional correctly identified odor). Right columns: stricter dichotomization at PST ≤4 (more conservative threshold). q-values are BH-FDR-corrected within outcome.

Outcome	Continuous PST β	Continuous PST 95% CI	Continuous PST q	Stricter PST≤4 β	Stricter PST≤4 95% CI	Stricter PST≤4 q
Mean MIMS/day	-0.501	(-0.827, -0.175)	0.077	-1.422	(-2.694, -0.150)	0.187
Sedentary min/day	-4.144	(-19.23, 10.94)	0.723	5.883	(-17.75, 29.51)	0.723
MVPA min/day	-2.897	(-6.136, 0.342)	0.222	-7.395	(-13.83, -0.960)	0.187
IS	-0.025	(-0.051, -0.0004)	0.187	-0.049	(-0.123, 0.026)	0.293
IV	0.038	(-0.0045, 0.081)	0.222	0.114	(-0.038, 0.266)	0.228
RA	-0.0046	(-0.014, 0.0052)	0.492	-0.022	(-0.048, 0.0041)	0.228
Total sleep (min)	-1.818	(-13.15, 9.518)	0.745	-9.075	(-29.95, 11.80)	0.508
Sleep efficiency	-0.0073	(-0.017, 0.0024)	0.228	0.0032	(-0.012, 0.018)	0.743
WASO (min)	2.734	(-0.753, 6.222)	0.228	-1.266	(-8.308, 5.776)	0.745
ASTP	0.0078	(-0.0017, 0.017)	0.228	0.037	(0.0009, 0.073)	0.187

Table S5. Multiple imputation (MICE) vs complete-case

OD effect estimates after multiple imputation by chained equations ($m = 10$), pooled via Rubin's rules with Barnard-Rubin degrees of freedom. CC = complete-case (right columns) for comparison. $n = 2,327$ or 1,903.

Outcome	MICE β	MICE 95% CI	MICE p	CC β	CC 95% CI	CC p
Mean MIMS/day	-0.57	(-0.900, -0.239)	<.001	-0.612	(-0.930, -0.294)	<.001
Sedentary min/day	-2.389	(-15.05, 10.27)	.712	-2.934	(-16.56, 10.69)	.663
MVPA min/day	-2.687	(-5.526, 0.153)	.064	-3.44	(-6.449, -0.431)	.026
IS	-0.028	(-0.048, -0.0077)	.007	-0.028	(-0.053, -0.0040)	.024
IV	0.05	(0.018, 0.082)	.003	0.048	(0.011, 0.084)	.012
RA	-0.0085	(-0.016, -0.0012)	.023	-0.0067	(-0.015, 0.0018)	.117
Total sleep (min)	-2.792	(-12.92, 7.331)	.589	-2.753	(-12.74, 7.235)	.577
Sleep efficiency	-0.0078	(-0.016, -0.0001)	.047	-0.0059	(-0.015, 0.0036)	.215
WASO (min)	2.942	(-0.051, 5.935)	.054	2.219	(-1.211, 5.649)	.196
ASTP	0.012	(0.0026, 0.022)	.013	0.011	(0.0030, 0.020)	.009

Table S6. Activity cut-point sensitivity

MVPA effects under Karas et al. (primary, ≥ 37.5 MIMS/min) vs Belcher et al. (alternative) cut-points for accelerometer state classification.

Cut-point set	Category	n	β	95% CI	p
karas	sedentary	2,055	-2.934	(-16.56, 10.69)	.663
karas	light	2,055	-33.8	(-47.96, -19.65)	<.001
karas	mvpa	2,055	-3.44	(-6.449, -0.431)	.026
wolff_hughes	sedentary	2,055	0.756	(-12.02, 13.53)	.904
wolff_hughes	light	2,055	-23.1	(-35.95, -10.25)	<.001
wolff_hughes	mvpa	2,055	-17.83	(-26.31, -9.354)	<.001
belcher	sedentary	2,055	-5.429	(-19.85, 8.988)	.447
belcher	light	2,055	-31.37	(-44.82, -17.92)	<.001
belcher	mvpa	2,055	-3.377	(-6.362, -0.392)	.028

Table S7. IPSW plug-in estimates

Inverse probability of selection weighting (IPSW) point estimates, treating the selection-model weights as fixed (no propagated uncertainty). MEC = primary survey-weighted estimate. n = 2,055.

Outcome	MEC β	MEC SE	MEC p	IPSW β	IPSW SE	IPSW p	% change
Mean MIMS/day	-0.612	0.156	<.001	-0.587	0.161	0.001	+4.06%
MVPA min/day	-3.44	1.471	.026	-3.436	1.486	0.028	+0.10%
IS	-0.028	0.012	.024	-0.027	0.012	0.033	+5.95%
IV	0.048	0.018	.012	0.043	0.018	0.02	-8.57%
ASTP	0.011	0.0041	.009	0.011	0.0042	0.015	-5.49%

Table S8. IPSW with weight trimming

IPSW with extreme weights trimmed at increasingly stringent percentiles (95th, 97.5th, 99th, 99.9th). 'Cap' = trimmed weight ceiling; 'n affected' = number of trimmed observations; '% change' = percent change in coefficient vs no-trim IPSW.

Percentile	Cap	n affected	MIMS % chg	MVPA % chg	IS % chg	IV % chg	ASTP % chg
0.95	1.307	103	+4.04%	+0.12%	+5.87%	-8.40%	-5.50%
0.975	1.32	52	+4.08%	+0.15%	+5.93%	-8.53%	-5.51%
0.99	1.332	21	+4.06%	+0.10%	+5.95%	-8.57%	-5.49%
0.999	1.355	3	+4.04%	+0.08%	+5.97%	-8.58%	-5.47%

Table S9. Bootstrap-IPSW with proper variance propagation

Cluster bootstrap ($B = 200$, PSU resampling within strata) propagating selection-model uncertainty into the primary outcome SEs. The SE-ratio column shows how much wider the bootstrap SE is vs the plug-in SE, indicating the cost of acknowledging selection-model uncertainty.

Outcome	Plug-in β	Plug-in SE	Bootstrap β	Bootstrap 95% CI	Bootstrap SE	SE ratio (boot/plug)
Mean MIMS/day	-0.587	0.161	-0.59	(-1.362, -0.059)	0.365	2.268
MVPA min/day	-3.436	1.486	-3.454	(-8.923, 3.677)	3.365	2.264
IS	-0.027	0.012	-0.026	(-0.060, 0.011)	0.018	1.548
IV	0.043	0.018	0.043	(-0.018, 0.112)	0.035	1.977
ASTP	0.011	0.0042	0.011	(-0.0044, 0.029)	0.0088	2.112

Table S10. Self-reported (CSQ240) vs PST-defined OD

Side-by-side comparison of M4 OD effects on each primary outcome using self-reported smell problem vs objective PST-defined OD as the exposure.

Outcome	Objective OD β (95%CI), p	Self-report β (95%CI), p	Objective sig?	Self-report sig?
Mean MIMS/day	-0.612 (-0.930, -0.294), p=0.000	-0.422 (-0.844, +0.001), p=0.050	YES	no
MVPA min/day	-3.440 (-6.449, -0.431), p=0.026	-6.022 (-8.932, -3.112), p=0.000	YES	YES
IS	-0.028 (-0.053, -0.004), p=0.024	-0.001 (-0.021, +0.019), p=0.920	YES	no
IV	+0.048 (+0.011, +0.084), p=0.012	-0.002 (-0.036, +0.032), p=0.905	YES	no
ASTP	+0.011 (+0.003, +0.020), p=0.009	+0.003 (-0.009, +0.015), p=0.602	YES	no

Table S11. Hour-by-hour OD effect on mean MIMS with BH-FDR

Survey-weighted OD effect on hourly mean MIMS at each of the 24 hourly windows (00:00 through 23:00). q -values BH-FDR-corrected across all 24 hours.

Hour	β	95% CI	p	q	Sig
00:00	-0.213	(-0.549, 0.124)	.206	.247	
01:00	0.133	(-0.212, 0.478)	.436	.455	
02:00	0.163	(-0.144, 0.470)	.287	.320	
03:00	0.092	(-0.218, 0.402)	.547	.547	

Hour	β	95% CI	p	q	Sig
		0.402)			
04:00	-0.215	(-0.626, -0.196)	.293	.320	
05:00	-0.702	(-1.285, -0.118)	.020	.027	*
06:00	-1.4	(-2.121, -0.678)	<.001	<.001	***
07:00	-1.818	(-2.463, -1.173)	<.001	<.001	***
08:00	-1.825	(-2.475, -1.175)	<.001	<.001	***
09:00	-1.623	(-2.314, -0.933)	<.001	<.001	***
10:00	-1.87	(-2.606, -1.134)	<.001	<.001	***
11:00	-1.654	(-2.313, -0.996)	<.001	<.001	***
12:00	-1.619	(-2.275, -0.963)	<.001	<.001	***
13:00	-1.282	(-1.944, -0.619)	<.001	<.001	***
14:00	-1.761	(-2.480, -1.042)	<.001	<.001	***
15:00	-1.781	(-2.454, -1.107)	<.001	<.001	***
16:00	-1.821	(-2.448, -1.194)	<.001	<.001	***
17:00	-2.276	(-3.013, -1.539)	<.001	<.001	***
18:00	-2.284	(-3.110, -1.458)	<.001	<.001	***
19:00	-1.988	(-2.833, -1.143)	<.001	<.001	***
20:00	-1.624	(-2.374, -0.874)	<.001	<.001	***
21:00	-1.284	(-1.931, -0.636)	<.001	<.001	***
22:00	-0.884	(-1.521, -0.247)	.008	.012	*
23:00	-0.522	(-0.974, -0.070)	.025	.032	*

Table S12. Effect modification: 20 OD × modifier interactions

Interaction *p*-values for OD × {age ≥60, sex, smoking, depression PHQ-9 ≥10} for each of the 5 primary outcomes. Significant interactions (*p* < 0.05) indicate effect modification. *p* < 0.05, ***p* < 0.01**, *p* < 0.001.

Outcome	Age (≥60) <i>p</i>	Sex (female) <i>p</i>	Smoking (ever) <i>p</i>	Depression (PHQ-9 ≥10) <i>p</i>
Mean MIMS/day	.016*	.543	0.231	0.095
MVPA min/day	.428	>.999	0.805	0.749
IS	.193	.840	0.638	0.448
IV	.025*	.226	0.47	0.918
ASTP	.043*	.108	0.369	0.086

Table S13. Olfactory subtype falsification test (food vs warning vs household)

M4 effects of each subtype-deficit indicator on each primary outcome. Subtypes: Food (chocolate, strawberry, grape, onion); Warning (smoke, natural gas); Household (leather, soap). *q*-values BH-FDR-corrected across 15 subtype × outcome contrasts. *n* = 2,055.

Outcome	Subtype	β	95% CI	d	<i>p</i>	<i>q</i>
Mean MIMS/day	Food	-0.155	(-0.569, 0.259)	-0.045	0.451	0.676
Mean MIMS/day	Warning	-0.145	(-0.588, 0.297)	-0.042	0.507	0.692
Mean MIMS/day	Household	-0.387	(-0.740, -0.034)	-0.113	0.033	0.225
MVPA min/day	Food	-0.952	(-6.585, 4.682)	-0.027	0.732	0.785
MVPA min/day	Warning	0.391	(-3.714, 4.497)	0.011	0.847	0.847
MVPA min/day	Household	-1.351	(-4.553, 1.850)	-0.038	0.395	0.662
IS	Food	-0.012	(-0.033, 0.0081)	-0.09	0.225	0.481
IS	Warning	-0.021	(-0.042, -0.0005)	-0.152	0.045	0.225
IS	Household	-0.015	(-0.038, 0.0078)	-0.109	0.188	0.469
IV	Food	0.012	(-0.033, 0.056)	0.052	0.59	0.702
IV	Warning	0.051	(0.0073, 0.095)	0.224	0.024	0.225
IV	Household	0.024	(-0.0051, 0.053)	0.104	0.103	0.31
ASTP	Food	0.002	(-0.0060, 0.010)	0.029	0.609	0.702
ASTP	Warning	0.004	(-0.0055, 0.013)	0.057	0.397	0.662
ASTP	Household	0.0068	(-0.0009, 0.014)	0.097	0.081	0.304

Table S14. Minute-level fragmentation phenotype

OD effects on minute-level PAXMIN_H fragmentation metrics: transition probabilities (ASTP, SATP), bout-length distributions, heavy-tail prolonged-sitting exposure, and time-of-day decomposition. *n* = 2,055, 2,051, 1,952, 1,948, 1,295, 684.

Metric	Label	β	95% CI	d	p	q
ASTP_minute	ASTP (active→sedentary)	0.012	(0.0043, 0.019)	0.174	.003	0.005
SATP_minute	SATP (sedentary→active)	-0.011	(-0.018, -0.0041)	-0.204	.003	0.005
transitions_per_wake_hour	Transitions per wake hour	-0.173	(-0.505, 0.159)	-0.091	.296	0.296
act_bout_mean	Active bout mean (min)	-0.158	(-0.398, 0.081)	-0.09	.186	0.248
act_bout_median	Active bout median	-0.044	(-0.124, 0.036)	-0.093	.270	0.309
act_bout_p90	Active bout p90	-0.446	(-1.059, 0.167)	-0.099	.147	0.235
act_bout_max	Active bout max	-10.1	(-16.35, -3.848)	-0.187	.003	0.007
sed_bout_mean	Sedentary bout mean (min)	0.295	(0.148, 0.442)	0.259	<.001	0.002
sed_bout_median	Sedentary bout median	0.098	(-0.0100, 0.207)	0.19	.074	0.147
sed_bout_p90	Sedentary bout p90	0.841	(0.385, 1.297)	0.263	<.001	0.003
sed_bout_max	Sedentary bout max	1.166	(-2.417, 4.749)	0.052	.511	0.511
frac_sed_in_long_bouts	Frac sed minutes in bouts ≥30 min	0.013	(0.0024, 0.024)	0.18	.019	0.037
frac_sed_in_very_long_bouts	Frac sed minutes in bouts ≥60 min	0.0053	(-0.0002, 0.011)	0.192	.058	0.058
sed_hazard_at_1min	sed hazard at 1min	-0.012	(-0.024, -4.03e-05)	-0.203	.049	0.189
act_hazard_at_1min	act hazard at 1min	0.003	(-0.0066, 0.013)	0.056	.523	0.581
sed_hazard_at_5min	sed hazard at 5min	-0.0061	(-0.013, 0.0013)	-0.137	.102	0.204
act_hazard_at_5min	act hazard at 5min	0.01	(0.0017, 0.019)	0.196	.021	0.189
sed_hazard_at_15min	sed hazard at 15min	-0.0068	(-0.018, 0.0041)	-0.097	.214	0.306
act_hazard_at_15min	act hazard at 15min	0.026	(-0.0020, 0.054)	0.339	.067	0.189
sed_hazard_at_30min	sed hazard at 30min	0.011	(-0.021, 0.044)	0.07	.474	0.581

Metric	Label	β	95% CI	d	p	q
act_hazard_at_30min	act hazard at 30min	-0.016	(-0.037, 0.0053)	-	.137	0.22
				0.129		9
sed_hazard_at_60min	sed hazard at 60min	0.084	(-0.0093, 0.178)	0.372	.076	0.18
						9
act_hazard_at_60min	act hazard at 60min	-0.007	(-0.035, 0.020)	-	.605	0.60
				0.04		5
				4		
astp_morn	ASTP morning (06-12)	0.015	(0.0061, 0.024)	0.19	.002	0.00
						5
astp_aft	ASTP afternoon (12-18)	0.014	(0.0056, 0.023)	0.185	.002	0.00
						5
astp_eve	ASTP evening (18-24)	0.015	(0.0065, 0.024)	0.192	.001	0.00
						5
satp_morn	SATP morning (06-12)	-0.012	(-0.020, -0.0032)	-	.009	0.00
				0.181		9
satp_aft	SATP afternoon (12-18)	-0.013	(-0.022, -0.0045)	-	.005	0.00
				0.214		6
satp_eve	SATP evening (18-24)	-0.011	(-0.019, -0.0037)	-	.005	0.00
				0.177		6

Table S15. Frailty-adjusted M4 sensitivity analysis

M4 estimates after adding (a) self-rated health (HUQ010) only, and (b) a multi-component frailty index.
Attenuation column = $(1 - \beta_{full}/\beta_{M4}) \times 100$.

Outcome	M4 β	M4 95% CI	M4+S RH β	M4+SRH 95% CI	M4+full frailty β	M4+full frailty 95% CI	Attenuatio n %
Mean MIMS	- 0.61 2	(-0.930, - 0.294)	-0.61	(-0.943, - 0.277)	-0.621	(-0.972, -0.271)	-150.0%
MVPA min/day	- 3.44	(-6.449, - 0.431)	-3.331	(-6.306, - 0.356)	-3.385	(-6.334, -0.436)	160.0%
IS	- 0.0 28	(-0.053, - 0.0040)	-0.029	(-0.053, - 0.0040)	-0.029	(-0.053, -0.0050)	-240.0%
IV	0.0 48	(0.011, 0.084)	0.048	(0.011, 0.085)	0.049	(0.012, 0.086)	-270.0%
ASTP per- person	0.01 1	(0.0030, 0.020)	0.011	(0.0030, 0.020)	0.012	(0.0030, 0.021)	-230.0%

Table S16. Counterfactual mediation through depression (PHQ-9) and BMI

Imai-Keele-Tingley natural effects decomposition. TE = Total Effect; NDE = Natural Direct Effect; NIE = Natural Indirect Effect (through mediator). p-value tests NIE = 0.

Mediator	Outcome	n	TE	NDE	NIE	p (NIE)
phq9_score	Mean MIMS/day	2,064	-0.678 (-0.974, -0.347)	-0.678 (-0.972, -0.320)	0.0003 (-0.028, 0.017)	<.001
phq9_score	MVPA min/day	2,064	-3.998 (-7.031, -1.421)	-4.002 (-7.157, -1.362)	0.0041 (-0.247, 0.199)	<.001
phq9_score	IS	2,064	-0.030 (-0.050, -0.0066)	-0.030 (-0.050, -0.0061)	3.49e-05 (-0.0019, 0.0015)	<.001
phq9_score	IV	2,064	0.049 (0.013, 0.080)	0.049 (0.012, 0.080)	-2.72e-05 (-0.0015, 0.0017)	<.001
phq9_score	ASTP	2,064	0.012 (0.0033, 0.020)	0.012 (0.0033, 0.020)	-4.89e-06 (-0.0004, 0.0006)	<.001
bmi	Mean MIMS/day	2,146	-0.704 (-1.059, -0.244)	-0.629 (-0.910, -0.321)	-0.074 (-0.210, 0.097)	>.999
bmi	MVPA min/day	2,146	-4.106 (-6.888, -0.895)	-3.502 (-6.666, -0.660)	-0.603 (-1.783, 0.772)	>.999
bmi	IS	2,146	-0.031 (-0.054, -0.0019)	-0.029 (-0.050, -0.0034)	-0.0023 (-0.0071, 0.0025)	>.999
bmi	IV	2,146	0.052 (0.014, 0.090)	0.049 (0.013, 0.085)	0.0024 (-0.0019, 0.0099)	>.999
bmi	ASTP	2,146	0.013 (0.0026, 0.022)	0.012 (0.0035, 0.020)	0.0015 (-0.0022, 0.0041)	>.999

Table S17. Compositional analysis of 24-hour activity budget

Isometric log-ratio (ilr) coordinates between OD and non-OD groups for the 4-part composition (sedentary, light, MVPA, sleep).

Coordinate	Interpretation	β	95% CI	p	n
ilr_z1_wake_v_sleep	wake-time activity ÷ sleep time	-0.108	(-0.168, -0.047)	.001	1,683
ilr_z2_active_v_sed	active wake ÷ sedentary wake	-0.162	(-0.231, -0.093)	<.001	1,683
ilr_z3_mvpa_v_light	MVPA ÷ light intensity	-0.13	(-0.225, -0.034)	.009	1,683

Table S18. Discrete-time hazard analysis of state transitions

OD effect on conditional probability of state-change at 1, 5, 15, 30, and 60 minutes of continuous activity or sedentary state.

Outcome	Label	β	95% CI	p	d	n
astp	ASTP (active→sedentary trans prob)	0.011	(0.0030, 0.020)	.009	0.162	2,055
mean_active_bout_prox	Mean active bout length (min)	-0.144	(-0.406, 0.119)	.272	-0.082	2,055

Outcome	Label	β	95% CI	p	d	n
mean_sed_bout_proxy	Mean sedentary bout length (min)	0.297	(0.154, 0.440)	<.001	0.259	2,055
transitions_per_day	Total transitions per day	-5.823	(-9.241, -2.405)	.002	-0.31	2,055

Table S19. Race/ethnicity-stratified mean MIMS estimates

Stratum-specific OD effects on mean MIMS by self-reported race/ethnicity. Exploratory; small subgroups have wide intervals. Race/ethnicity strata defined per NHANES RIDRETH3.

Race/ethnicity	Outcome	n (total)	n (OD)	β	95% CI	p
NHWhite	Mean MIMS/day	1,010	127	-0.809	(-1.226, -0.392)	<.001
NHWhite	MVPA min/day	1,010	127	-3.76	(-7.168, -0.353)	.032
NHWhite	IS	1,010	127	-0.032	(-0.064, 0.0004)	.053
NHWhite	IV	1,010	127	0.045	(-0.0017, 0.092)	.058
NHWhite	ASTP	1,010	127	0.016	(0.0035, 0.028)	.013
NHBlack	Mean MIMS/day	438	81	-0.232	(-1.186, 0.721)	.622
NHBlack	MVPA min/day	438	81	-0.395	(-7.926, 7.137)	.915
NHBlack	IS	438	81	-0.0068	(-0.047, 0.033)	.731
NHBlack	IV	438	81	0.082	(0.0085, 0.156)	.030
NHBlack	ASTP	438	81	0.0029	(-0.016, 0.022)	.764
MexAm	Mean MIMS/day	240	35	0.515	(-1.244, 2.273)	.548
MexAm	MVPA min/day	240	35	7.766	(-21.24, 36.77)	.583
MexAm	IS	240	35	-0.021	(-0.084, 0.042)	.494
MexAm	IV	240	35	0.0053	(-0.121, 0.132)	.931
MexAm	ASTP	240	35	-0.0009	(-0.023, 0.021)	.928
OtherHisp	Mean MIMS/day	162	28	-2.62	(-4.502, -0.739)	.008
OtherHisp	MVPA min/day	162	28	-26.63	(-44.15, -9.105)	.004
OtherHisp	IS	162	28	-0.074	(-0.132, -0.016)	.014
OtherHisp	IV	162	28	0.144	(0.0048, 0.283)	.043
OtherHisp	ASTP	162	28	0.05	(0.011, 0.089)	.014
NHAsian	Mean MIMS/day	170	35	0.134	(-0.737, 1.006)	.754
NHAsian	MVPA min/day	170	35	-5.308	(-13.57, 2.956)	.198
NHAsian	IS	170	35	-0.024	(-0.075, 0.028)	.354
NHAsian	IV	170	35	-0.041	(-0.133, 0.052)	.376
NHAsian	ASTP	170	35	-0.013	(-0.032, 0.0074)	.206

Table S20. Selection bias: included vs excluded PST-eligible participants

Characteristics of the 2,327 included participants vs the 530 excluded for insufficient accelerometer wear, to assess selection bias.

Variable	Included (n=2,327)	Excluded (n=530)	p (difference)
age	58.7675	56.0283	<.001
sex	1.52127	1.48679	.166
od_b	0.132789	0.150943	.303
bmi	29.6372	28.4493	<.001
comorb_count	0.981092	0.707547	<.001
phq9_score	3.19506	2.51467	<.001
pir	2.68503	3.03388	<.001

Table S21. Stratified split-half replication

Random 50/50 split (preserving OD prevalence; seed 42) with independent M4 fit in each half. 'Direction-consistent' means both halves agree on sign; 'Both significant' means both halves reach $p < 0.05$.

Outcome	Full β	Full 95% CI	Half A β	Half A 95% CI	Half B β	Half B 95% CI	Direction-consistent	Both significant
Mean MIMS	-0.612	(-0.930, -0.294)	-0.796	(-1.358, -0.234)	-0.457	(-0.965, 0.052)	True	False
MVPA min/d	-3.44	(-6.449, -0.431)	-4.007	(-7.938, -0.076)	-3.194	(-9.335, 2.947)	True	False
IS	-0.028	(-0.053, -0.0040)	-0.028	(-0.061, 0.0059)	-0.025	(-0.053, 0.0036)	True	False
IV	0.048	(0.011, 0.084)	0.093	(0.036, 0.150)	-0.0036	(-0.063, 0.056)	False	False
ASTP (per-person)	0.0114	(0.0031, 0.020)	0.016	(0.0016, 0.030)	0.0074	(-0.0063, 0.021)	True	False
ASTP (minute-level)	0.0123	(0.0043, 0.019)	0.018	(0.0040, 0.032)	0.0063	(-0.0064, 0.019)	True	False
SATP (minute-level)	-0.011	(-0.018, -0.0040)	-0.01	(-0.021, 0.0012)	-0.012	(-0.021, -0.0038)	True	False

Figure S1. Hour-by-hour OD-non-OD activity gap with FDR significance
(top: 24-hour profile; bottom: hourly OD effect with BH-FDR across 24 tests)

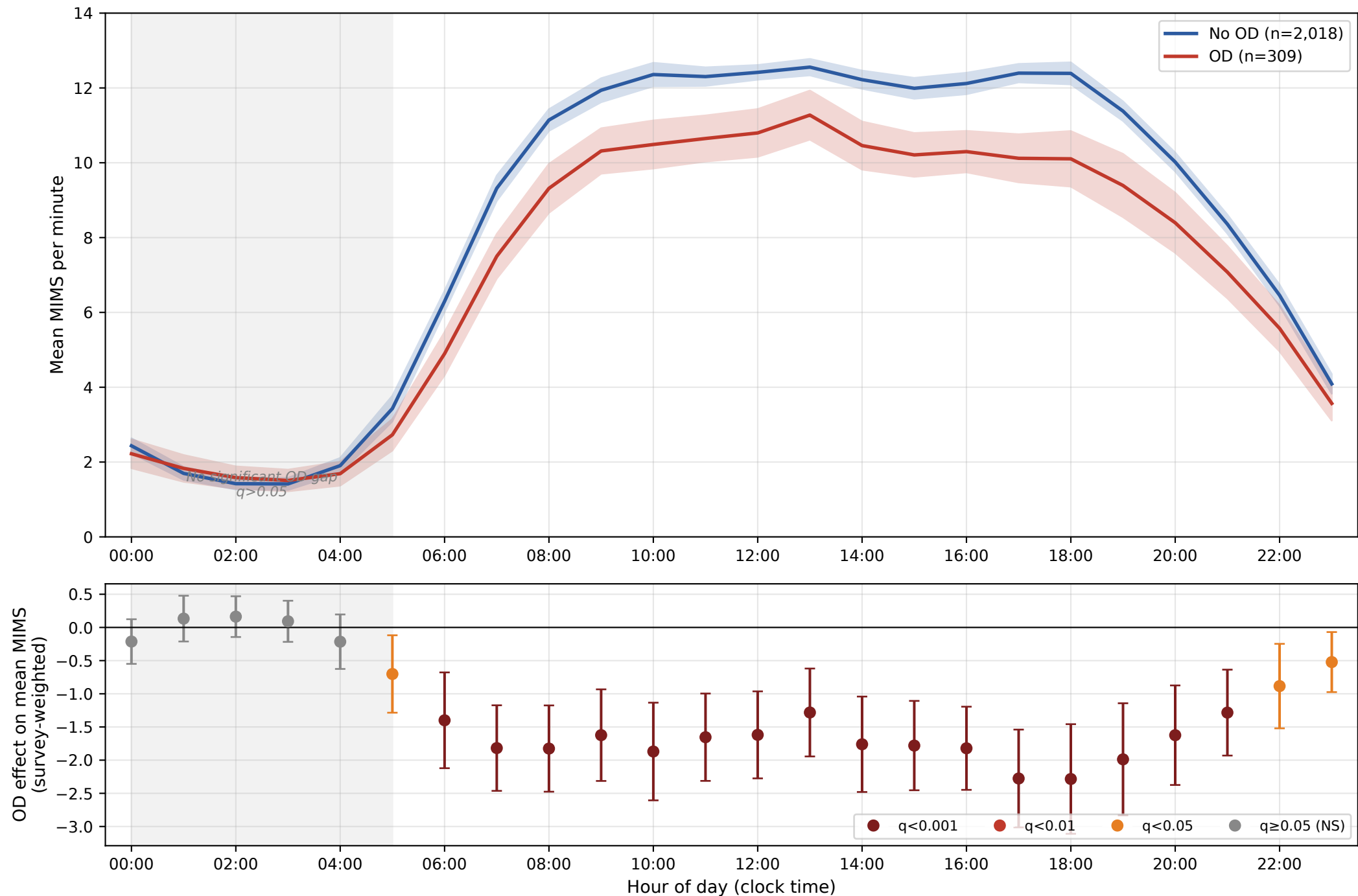
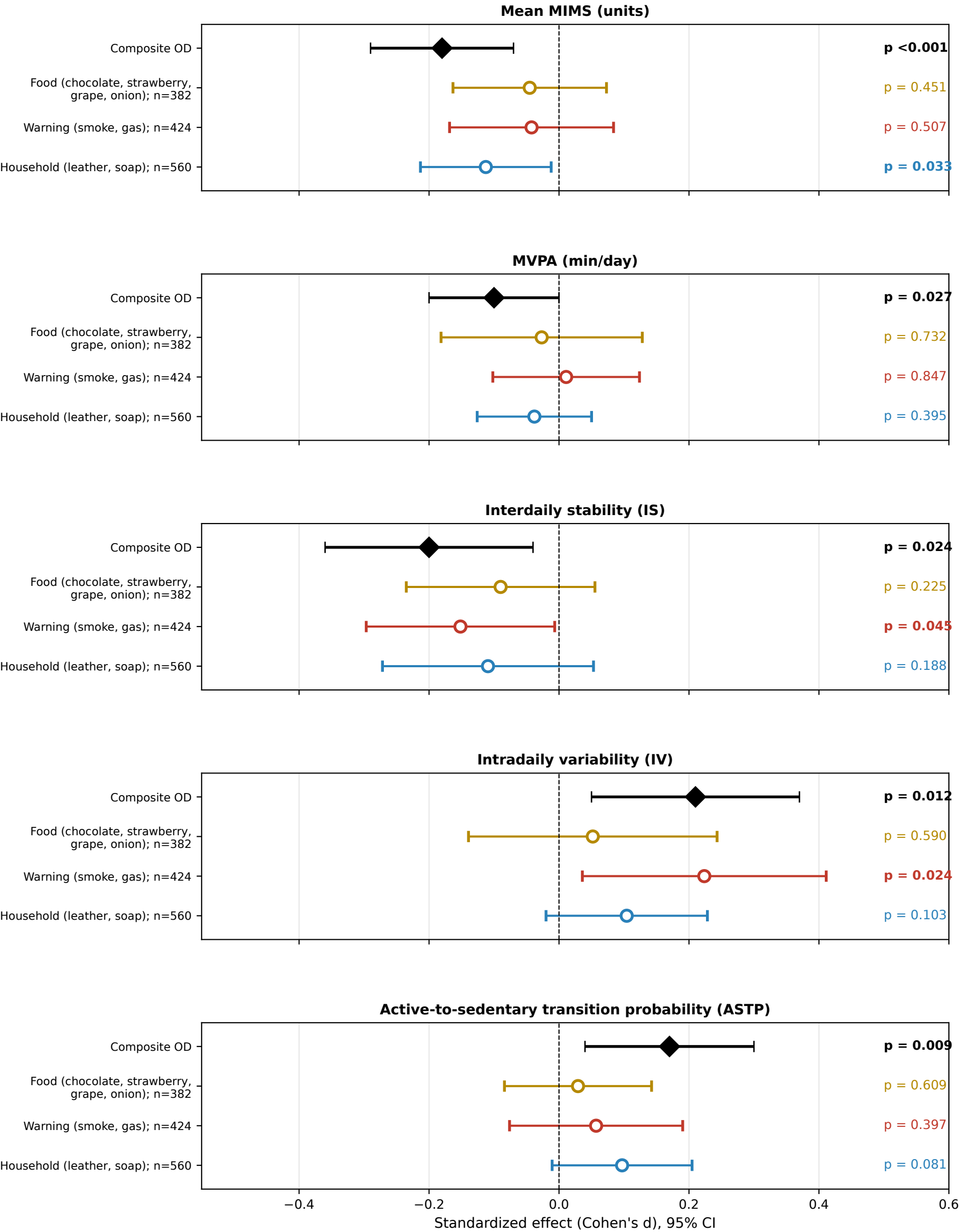


Figure S2. Olfactory dysfunction subtype effects on activity outcomes



M4-adjusted estimates. Composite OD (top row of each panel, black diamond) is the primary exposure (PST ≤5 of 8 correct). Food, warning, and household subtype deficits are mutually exclusive indicators within each odor category. After BH-FDR correction across the 15 subtype × outcome contrasts, no individual subtype contrast survives $q < 0.05$ (smallest $q = 0.225$). Subtype results are reported as exploratory; bold p-values are $p < 0.05$ uncorrected.